

Free-Body Diagrams (FBDs)

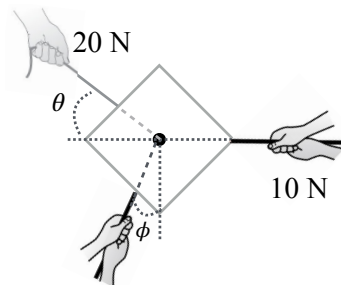
A FBD is a picture that illustrates all of the forces acting on a particular object. The object is “free” from its environment, as only the forces are shown. FBDs are essential in Mechanics and will be **required** for all of your solutions to Dynamics (Newton’s Laws) problems.

Steps:

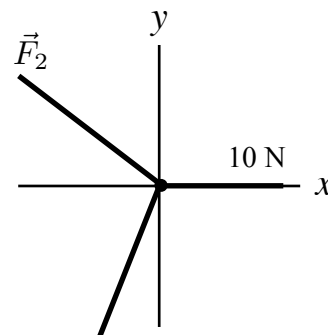
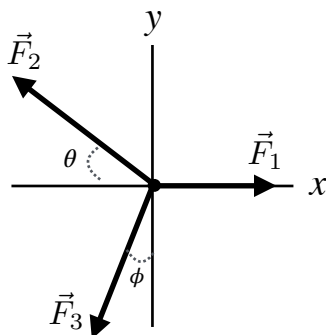
1. Isolate the object (system) being analyzed.
Identify ALL forces that act ON the object, NOT the forces the object exerts on its surroundings.
2. Draw a *convenient* coordinate system.
3. Represent the object as a dot at the origin, when possible (particle model).
4. Draw **vectors** representing each of the identified forces.
 - Place the tail of each force vector on the object.
 - Illustrate the angles with respect to the coordinate axes.

Free-Body Diagram Sketching: Dos and Don’ts

Example: top view of a box pulled by three ropes.



FBDs:



✓ DOs:

- Represent forces with vectors (arrows)
- Label each vector with a symbol
- Illustrate the angles wrt the coordinate axes
- Place the tail of the vector on the object (dot)

✗ DON'Ts:

- Represent forces with lines without arrowheads
- Label vectors with values, or leave unlabeled
- Forget to identify the angles
- Place the tail of the vector at random locations

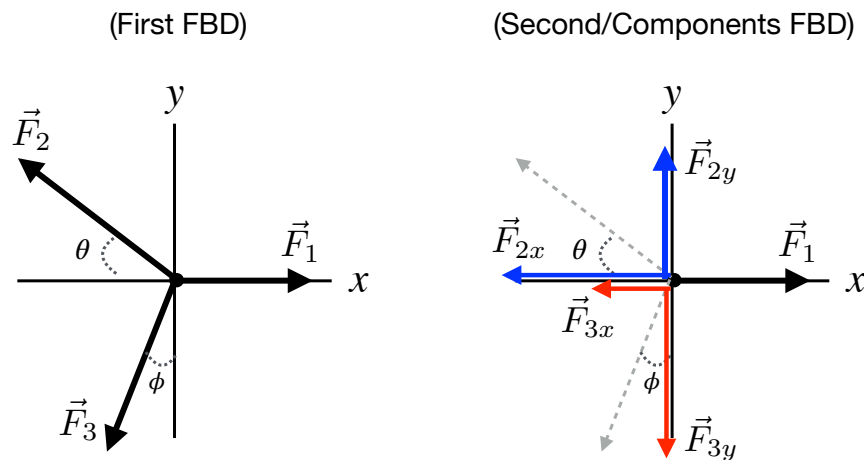
Second (Components) Free-Body Diagram

If there are vectors not aligned with the axes in the first FBD, draw a second FBD where the non-aligned vectors are decomposed (“broken”) along said axes.

This Second—or Components—FBD is a like-copy of the first one:

- it must follow the same orientation of the axes; and
- it must include all force vectors.

The only difference between the two FBDs is that the latter illustrate the *component vectors* of the non-aligned forces; all vectors now point exactly along the x- or y-axes.



Component Vectors and Components:

After sketching the Components FBD, determine the magnitudes of the component vectors in terms of the original vector magnitudes and functions of the angle(s). This requires using trigonometry.

For example:

$$F_{2y} = F_2 \sin \theta$$

$$F_{3y} = F_3 \cos \phi$$

Note:

- $\vec{F}_{2x}$ and $\vec{F}_{2y}$ are called the “component vectors” of $\vec{F}_2$, while their magnitudes, F_{2x} and F_{2y} , are simply referred to as the “components”.
- The trigonometric relationships only hold for the magnitudes of the original vectors and vector components, not the vectors themselves. Do not cap these vector magnitudes with arrows, for instance, $\vec{F}_{2x} \neq F_{2x}$.